DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Multi-Turn DPO for Tool-Augmented LLMs | arXiv:2504.02882

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must control conversation flow by deciding whether to ask follow-up questions, make a tool call, or reject inappropriate requests. SFT with expert trajectories alone cannot learn these conversational skills, leading to **hallucinated tool invocations**.

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Missing Follow-ups

Failing to ask follow-up questions when user queries lack required arguments

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Hallucinated Calls

Making tool invocations with incomplete or fabricated parameter values

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Failed Rejection

Unable to reject out-of-scope requests when no suitable tool exists

Modeling TA-LLM Dialogue as a 5-State MDP

S1
Initial
No history. Out-of-scope requests return rejection.

→

S2
Tool Selected w/o Slots
Tool identified but required parameters missing.

→

S3
Tool Selected w/ Slots
Tool and all argument values determined.

→

S4
Wait for Response
Tool call executed, awaiting results.

→

S5
Complete
Tool results received, completion generated.

Five internal (unobserved) states govern all TA-LLM interaction dynamics

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example Scenario
Type 1	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.	"Book a flight from NYC to LAX on March 15"
Type 2	1 → (2×N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times.	"Book a flight to LAX" → "What date?" → "March 15"
Type 3	1 → 1	Requested functionality is unavailable. Tool call must be rejected.	"What's the weather on Mars?" → Rejection message

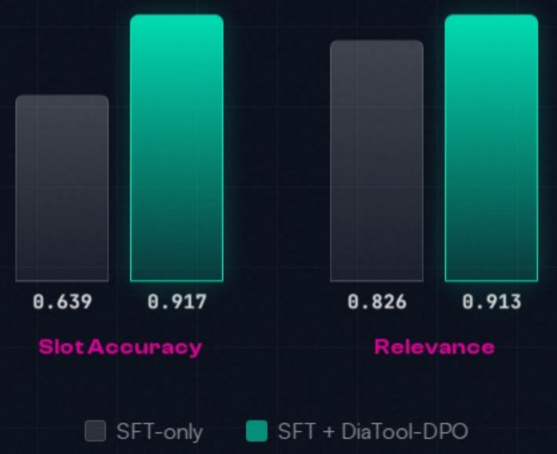
Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen	Rejected	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot and +10.5% Relevance with DiaTool-DPO



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

01 **MDP formulation enables systematic preference pair construction** — Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

02 **DiaTool-DPO consistently improves Macro Average** — Across all three model sizes (3.1B, 8B), SFT + DiaTool-DPO achieves the best balanced performance on FunctionChat-Bench.

03 **Slot-filling and tool rejection see the largest gains** — LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only.

04 **DPO alone is insufficient** — DiaTool-DPO-only (without SFT) performs poorly, confirming that DPO complements rather than replaces supervised training.

05 **The approach is language-agnostic** — While primary experiments were in Korean, the method generalizes to English, opening broad applicability for production TA-LLMs.

DiaTool-DPO opens the path to GPT-4o-level conversational tool use on smaller open-source models